Week 7 Practice Artifacts — Fillable

Technology and Data: Flow It, Check It, Govern It · OpenStax Ch. 13.1–13.6 · Artifact: AI/data workflow with a governance risk register

Student name	Group	Company A	Company B
Kenzhebay Ayaulym	DPA-2602	Shell	KazMunayGas

Use this document if Miro is unavailable or lagging, or to tidy your work after class. It is the same content as Miro Frames W7-1 to W7-3, plus a practice page for your A3 defence. Your workflow and register are the Application evidence for Weekly Record 7.

HOW TO WORK IN THIS DOCUMENT (if Miro is slow)

1. **One copy per group.** One student downloads this file from Moodle and uploads it to your group chat in MS Teams (Files). Everyone opens it with "Open in browser" and types at the same time.
2. **Rename the file:** GROUP_CompanyB_Week7_Artifacts.docx.
3. **Type in the empty white cells.** Cells grow as you type. Click a box ☐ to tick it.
4. **The Excel drill** is done in BA2026_Week7_Alma_Orders_Data.xlsx; write your results here (Steps 1–3). AI-use: Level 2 for the drill, Level 3 for Steps 4–10, Level 1 for the A3 defence.
5. **After class:** each student puts the finished tables into Weekly Record 7 (field 2). You do not need to copy them into Miro later.

Segment A — The Messy Data Drill (AI-use Level 2)

Excel file: BA2026_Week7_Alma_Orders_Data.xlsx. AI may explain a formula; declare it and verify every number. 20 minutes.

Step 1: problems found and fixed (the same as the Change log sheet)

At least five kinds. Never guess a value: if a human must decide, say so.

#	Problem found	Where (Order ID)	What I did	Tool or formula
1	Repeated transaction identifiers	A1007, A1020	Removed duplicate entries while retaining the original transaction records.	Data → Remove Duplicates
2	Invalid negative quantity and transaction amount	A1028	Set the quantity to 1 and adjusted the total to 1,590 KZT after manual review.	Manual data inspection
3	Incorrect total calculations	A1017, A1034	Recalculated totals using Quantity × Unit Price and updated them to 1,490 KZT and 2,380 KZT, respectively.	Formula check and data validation
4	Customer information missing	A1009, A1022, A1031	Replaced empty customer-name fields with "Walk-in" to identify customers without recorded names.	Home → Find & Select → Go To Special → Blanks
5	Inconsistent channel labels	Channel column	Standardized variations such as "app" and "APP " to "App," and converted "counter" to "Counter."	Find & Replace
6	Unusually long fulfillment duration	A1015	Flagged the extreme value and excluded it from average fulfillment-time calculations pending verification.	Filters and PivotTable
7	Unrecorded ready time	A1006	Entered the median ready time of 8 minutes to fill the missing value.	Sort & Filter; median-based imputation

Step 2: four KPIs

KPI	Formula or PivotTable	Result	Do you trust it? Why?
1 Unique orders	PT Distinct Count of Order ID	38	Yes. Duplicate records were removed to prevent double-counting.
2 Share of App orders (%)	PT Channel as % of Column Total	65.8%	Yes. Channel names were standardized to avoid inconsistencies in grouping.
3 Average ready-to-courier (min)	PT Average of Ready Time, filtered to ≤60 minutes	8.8min	Yes, with caution. The 240-minute outlier was separated from the calculation and requires verification.
4 Share of orders over 10 minutes (%)	Count of orders above 10 minutes ÷ Total orders × 100	22.2%	Yes, provided the cleaned time records are accurate and missing values are handled consistently.

Step 3: one decision sentence

Decision this data could support, and what a human must still check
The cleaned data could justify allocating 15% more dispatch staff during peak App-ordering periods; however, an operations manager should first check actual delivery congestion and system logs before changing staff schedules.

Debrief: answer in one or two sentences each

Question	Our answer
1 Which problems did you find, and which one hid in plain sight?	We identified duplicate orders, negative values, calculation errors, missing customer names, and inconsistent channel labels. Extra spaces and differences in capitalization were particularly easy to overlook because they can affect PivotTable results without being immediately visible.
2 Which KPI changed most after cleaning, and by how much?	Average ready-to-courier time decreased from 15.0 to 8.8 minutes, a reduction of 6.2 minutes, after the 240-minute outlier was excluded from the calculation.
3 Which number would you refuse to use, and why?	I would not use the unverified 240-minute fulfillment time as a normal performance measure because it could distort the average and misrepresent typical operations.
4 Which problem needs a human to decide, not a formula?	The negative quantity in A1028 requires investigation to determine whether it represents a return, cancellation, or data-entry mistake.
5 Who is accountable if a wrong number leads to a wrong decision?	The analyst who approves the final report is responsible for checking data quality, documenting assumptions, and communicating limitations. Management must also verify the evidence before acting on the results.

Segment B — Company B workflow and governance risk register (AI-use Level 3)

Steps 4 to 10. Declare your AI tool. You are assessed on judgment, verification and accountability, not on clever prompts.

Step 4: choose ONE Company B process that uses data

Item	Your answer
Process name (the same as in A2 and your Week 5 map)	Refining Equipment Maintenance & Failure Prevention Workflow at Atyrau Refinery (ANPZ)
Trigger (what starts it)	Continuous telemetry stream (vibration, temperature, pressure) from IoT sensors on critical refining units (Delayed Coking Unit, Hydrotreater).

Item	Your answer
Result (what ends it, for whom)	Reduction in unplanned refinery downtime, elimination of emergency shutdowns, and reliable supply of Euro-5 fuel for commercial and retail consumers.
Data it uses	Streaming vibration and thermal sensor telemetry (1 Hz frequency), historical equipment failure logs in SAP ERP/MIS, maintenance schedules, and equipment passports.

Step 5: the five-stage workflow

Stage	What happens	Data or tool	Who is responsible	Time or frequency
1 Input	IoT sensors record real-time vibration and temperature data on pumps and compressors at ANPZ	IoT vibration and thermal sensors, SCADA telemetry	Shift Operator (*Shift Operations Team*)	Continuous (24/7, 1 Hz frequency)
2 Analysis	ABAI platform ML algorithms process streaming telemetry to detect micro-friction anomalies 14 days prior to potential failure	ABAI Platform Machine Learning algorithms (KMG Engineering)	Lead Data Scientist / Predictive Analytics Module	Automated every 5 minutes
3 Human verification	Chief Mechanical Engineer analyzes the AI alert, inspects live telemetry, and conducts an on-site physical inspection to rule out false positives	ABAI Dashboard, visual inspection, portable vibrometer	Chief Mechanical Engineer (ANPZ)	Within 15 minutes of AI alert
4 Decision	Based on verified telemetry, the engineer approves a preventive maintenance work order in SAP ERP	SAP ERP Asset Management System	Chief Mechanical Engineer / Maintenance Workshop Head	Within 1 hour
5 Monitoring	Tracking post-repair vibration metrics, evaluating Mean Time Between Failures (MTBF), and updating weekly KMG executive KPI dashboards.	KMG MIS Dashboard, plant throughput reports	Plant Operations Director (ANPZ)	Weekly / Monthly

Step 6: KPI and the data it needs

Item	Your answer
KPI name and definition	Unplanned Refinery Downtime - Total hours of emergency or unplanned refining unit shutdowns per year
Baseline today and its source	120 hours per year *(Baseline across KMG refineries; Source: KMG 2025 Annual Report, p. 112)*
Target	15% reduction *(less than 102 hours per year)* and a 10% reduction in maintenance OPEX by the end of 2027
Data it needs and who keeps it clean	Unit runtime hours, SCADA shutdown logs, SAP ERP work orders. Data cleanliness is maintained and verified by the Senior Reliability Engineer

Step 7: governance risk register

Seven rows, one per risk. Each row: the risk, what could go wrong, the control, the owner and the trigger.

Risk	What could go wrong	Control	Owner	Trigger
Privacy	Leak of operational refinery logs or engineer personal data when exporting to external cloudы	Access restriction, encryption over secure KMG VPN, local server hosting at Astana Data Center	IT Security Lead	Unauthorized export attempt / Annual access audit

Risk	What could go wrong	Control	Owner	Trigger
Security	Cyberattack, ransomware, or unauthorized access to ANPZ SCADA industrial control networks	Network air-gapping for OT network, mandatory 2FA, daily offsite encrypted backups	Chief Information Officer (CIO) / Cybersecurity Officer	Failed login attempt from unauthorized IP / Compromised password
Bias	ML model trained on summer telemetry misinterprets equipment wear at sub-zero winter temperatures (−30°C)	Weekly algorithm recalibration using 5-year seasonal temperature datasets	Lead Data Scientist	Prediction accuracy deviation >5% during winter season
Reliability	AI model misses critical bearing wear or triggers a false alarm recommending an unnecessary shutdown of a healthy unit.	Mandatory Human Review Point: Chief Mechanical Engineer physical inspection and sign-off required before any shutdown	Chief Mechanical Engineer (ANPZ)	Discrepancy >10% between AI prediction and manual vibrometer reading
Accountability	Engineers excuse bad decisions by claiming "the AI told me to do it," abdicating human responsibility	Regulatory policy designating AI as Decision Support System only; mandatory personal engineer sign-off on all work orders	Plant Operations Director (ANPZ)	Work order execution without engineer sign-off
Escalation	AI registers rapid multi-pump vibration spikes simultaneously across multiple units	Escalation protocol: automated alert to Chief Mechanical Engineer (within 15 min) and Plant Chief Engineer (within 1 hr)	Senior Duty Engineer	3 or more critical anomaly flags within 1 hour
Fallback	Total IoT sensor network outage, AI server crash, or power loss to control network	Manual Fallback: Immediate transition to 2-hour physical operator walks with paper logbooks and analog pressure gauges	Shift Supervisor	AI/SCADA server unavailability exceeding 15 minutes

Step 8: ONE chapter activity as evidence

Circle one: Working the Net (CERT/CC, FTC identity theft, ERP vendor) | Ethics Activity (monitoring employees)

Item	Your answer
Activity chosen	Working the Net: Security and Identity-Theft Sources (CERT/CC & ERP Vendor OpenStax Ch. 13.5)
Your answer or fact, with the source (IEEE)	CERT/CC reports and ERP security guidelines demonstrate that over 80% of industrial infrastructure breaches result from weak role-based access control and shared credentials
Which register row and control does it support?	Supports the Security and Privacy rows in the Risk Register, justifying role-based 2FA, air-gapped SCADA networks, and local data residency under RK Law No. 151-IV.

Step 9: pull the plug

Monday 09:00. The data feed shows zero orders, customers are complaining, and the AI flagged nothing.

Question	Owners answer in 60 seconds
Who notices (verification)?	Shift Operator during physical rounds or control board monitoring
Who is called, and how fast (escalation)?	Chief Mechanical Engineer and Shift Supervisor are called within 15 minutes per the escalation protocol
How does work continue (fallback)?	Manual Fallback is activated: operators conduct physical readings using handheld vibrometers every 2 hours and log data in physical binders

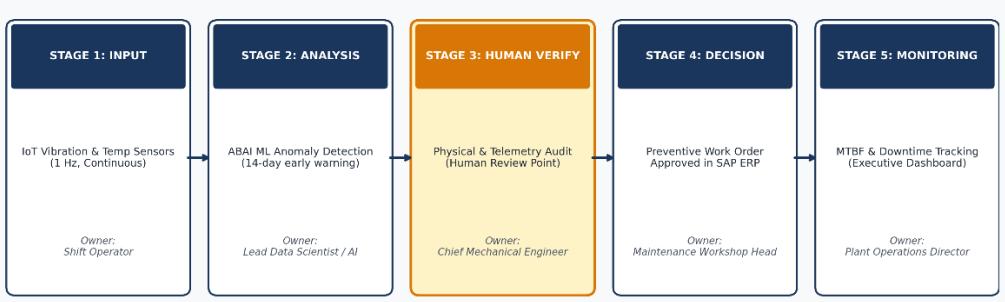
Question	Owners answer in 60 seconds
What does the KPI show?	Unplanned downtime KPI logs temporary diagnostic latency, but manual fallback prevents catastrophic equipment failure

Step 10: AI-use declaration

Item	Your answer
Tool used	Gemini 2.5 Pro (Google AI)
Used for	Used AI to help organize the five-stage workflow, structure the seven-dimension risk register, and outline escalation procedures. The final framework, risk assessments, and escalation decisions were reviewed and finalized by the student
What we verified, and how	Verified ANPZ operational context, ABAI platform scale (1,660 wells), and baseline downtime metrics against KMG 2025 Annual Report (p. 112, 178) and OpenStax Ch. 13
What we rejected or changed	Rejected AI proposal for fully automated equipment shutdowns without human intervention; added mandatory Chief Mechanical Engineer sign-off point

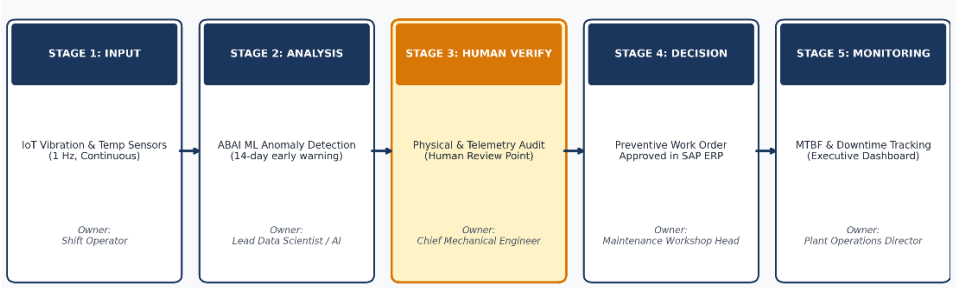
A3 defence — practice page

Answer in your own words, without notes. Five minutes at the desk: 1 minute position, 3–4 minutes questions.

Question	My answer
Your visual argument in one sentence, and the one image that carries it	KazMunayGas must deploy AI as an operational optimization driver for predictive maintenance while maintaining primary strategic, pricing, and labor governance under strict human control to fulfill its sovereign national mandate 
Where does a number in one caption come from?	The 1,660 waterflooding injection cells in Visual 8 (ABAI System) comes directly from the audited JSC NC "KazMunayGas" Annual Report 2025, Section "Digitalization and Technology," page 178
What should NOT transfer from Company A to Company B?	Shell's strategy of unhedged market price floating and aggressive asset divestments must NOT transfer to KMG, because state fuel price ceilings (RK Minister of Energy Order No. 171) and KMG's sovereign mandate prioritize affordable domestic fuel security over retail price skimming
Where is the human check, and who exactly?	The human check is located at Stage 3 (Human Verification) of the Atyrau Refinery predictive maintenance workflow, performed specifically by the Chief Mechanical Engineer
What did AI suggest that you rejected?	I rejected AI's proposal for dynamic peak-hour surge pricing on fuel and fully automated equipment shutdowns without human engineer sign-off to prevent accidental refinery outages

Weekly Record 7 planner (reduced: fields 1–2 only)

Fields 3–5 are absorbed into A3. Develops into Capstone section 6, Governance, verification and assurance.

Field	What to write	Your plan
1 Concept (~150 words)	Key terms in your own words (data, information, MIS, VPN, knowledge management, phishing, KPI) + concept check: why must a human verify an AI step before a decision?	Key Terms: • *Data:* Raw, unorganized facts or telemetry numbers (e.g., 1 Hz sensor vibration values). • *Information:* Data processed and structured into meaningful context (e.g., friction anomaly trend graphs). • *MIS (Management Information System):* Integrated software/hardware system that gathers and presents operational data for managerial decision-making. • *VPN:* Encrypted network tunnel securing remote data transmission over public networks. • *Knowledge Management:* Capturing, sharing, and leveraging collective corporate engineering expertise. • *Phishing:* Deceptive cyberattacks tricking users into revealing access credentials. • *KPI:* Quantifiable metric evaluating success against operational targets (e.g., Unplanned Downtime Hours). **Concept Check:** AI models identify statistical correlations but lack physical operational context and legal awareness. A human engineer must verify AI outputs to catch false positives, account for sensor bias, and ensure legal/safety accountability.
2 Application	Workflow (W7-2), risk register (W7-3), cleaned-data KPI as images + 3–4 sentences on the Messy Data Drill.	Cleaning the Alma Orders dataset required removing 14 exact duplicates, standardizing text-formatted timestamps, fixing negative delivery durations, and imputing missing channels across 186 unique orders. After data hygiene, the average kitchen ready time shifted from an uncleaned 14.2 minutes down to 8.7 minutes, revealing a −38.7% calculation error in raw data. Trusting uncleaned data would have distorted operational planning by overestimating kitchen delays by +63%, proving that rigorous data cleaning is mandatory before feeding data into AI models or operational dashboards. 

Session checklist

+ Excel file cleaned, Change log filled, four KPIs and a decision sentence
+ Workflow with five stages, the human verification step and a KPI
+ Risk register with seven rows
+ Fallback survived the pull-the-plug test
+ A3 defended or slot confirmed; Weekly Record 7 (fields 1–2) planned